

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_replay_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	41.81978	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	76.36324	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1396.33112	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	232.65356	1	0	0	0.948791	16.25
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	302.03618	1	0	0	0.96266	14.92
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.069352	0.069352	2193.00564	3	0	0	0.553475	15.72333

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 41.81978.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 76.36324.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1396.33112.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 232.65356.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 302.03618.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
 - X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
 - X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.061781.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.069352.
- Held-out runtime ms: 2193.00564.
- Accepted epoch count: 3.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002918.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.097465, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.093363, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.058952, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.052342, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.044637, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Aspect	Summary
Direct synthesis	direct_generate_plus_one_repair failed feasibility (0% heldout feasible rate) and had very high gaps.
Budget-matched independent search (no replay)	budget_matched_no_replay also failed feasibility (0%) with high penalized gap (3.0).
Replay-aware iterative search	replay_solver_evolution achieved perfect feasibility (100%) with the best heldout feasible gap (0.0694), beating baselines in gap but had highest runtime (2193 ms).
Fixed-baseline context	Baselines (nearest neighbor, Clarke-Wright, regret insertion) all had 100% feasibility; Clarke-Wright had best gap (0.0738), runtime from 42 to 1396 ms, regret insertion most expensive and worst gap.
Feasibility	Learned conditions without replay had 0% feasibility; with replay achieved 100%. Baselines 100%.
Objective gap	Replay solver evolution gap (0.0694) slightly better than best baseline Clarke-Wright (0.0738).
Runtime	Baselines fastest nearest neighbor (42 ms) and Clarke-Wright (76 ms); replay approach was slowest (2193 ms).

Aspect	Summary
Robustness across held-out families	Replay approach had low robustness dispersion (0.0029) similar to baselines, indicating stable performance.
Learned vs fixed baselines	No learned method without replay beat fixed baselines. Replay-aware iterative search slightly outperformed the best fixed baseline gap-wise.
Replay vs no replay control	Replay clearly outperformed budget-matched no-replay in feasibility and gap.

Conclusions

- The ****best heldout condition**** was the ****replay_solver_evolution****, achieving perfect feasibility and better gap than fixed baselines.
- ****Replay mechanism provided a significant advantage**** over budget-matched no-replay learning control.
- Learned methods without replay failed to generate feasible solutions.
- Baselines remain competitive in runtime but are slightly worse in gap compared to replay-based learning.
- The replay-aware iterative approach trades off runtime for improved solution quality and robustness.